# Demand-responsive transport in rural areas

Case study of the municipality of Boticas

Google Earth

# Why?

Congestion

Car dependency

*But in **rural areas**...*

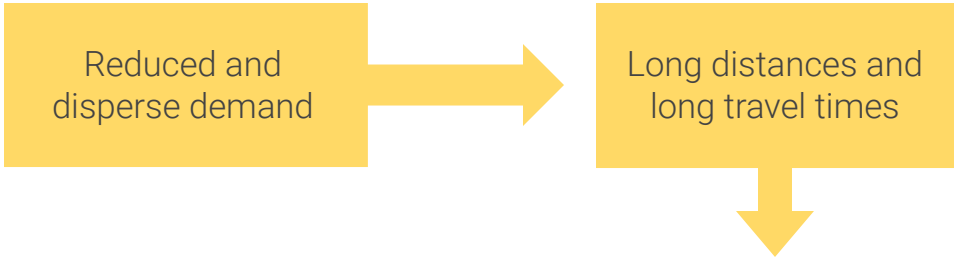
Congestion

Car dependency

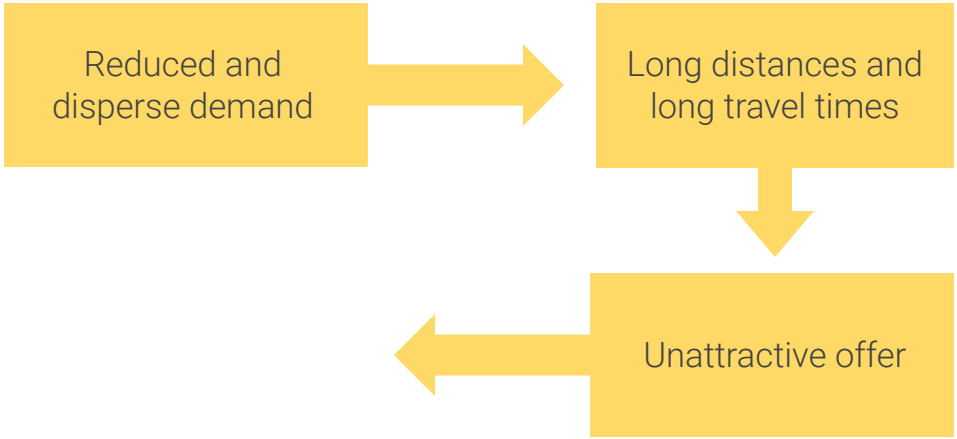
*In* ***rural areas...***

Reduced and disperse demand

*In* ***rural areas...***



*In* ***rural areas...***



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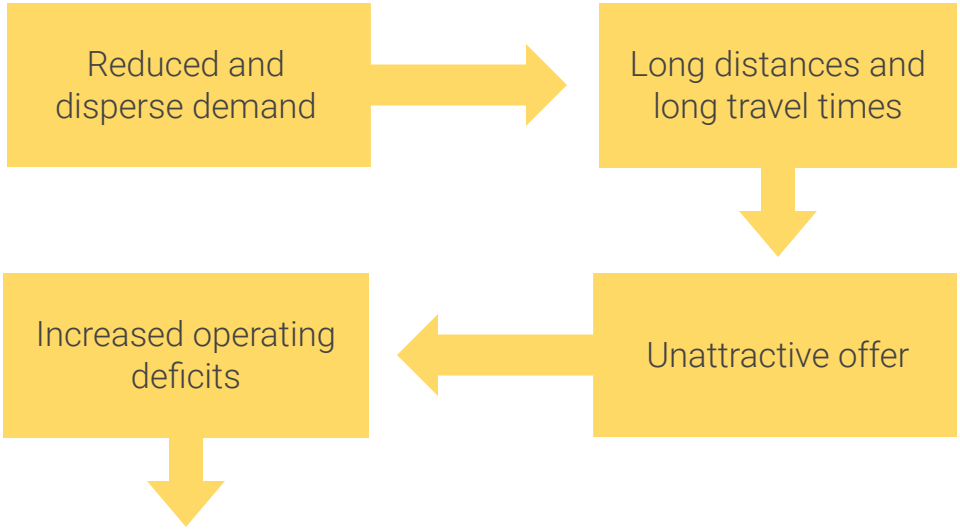

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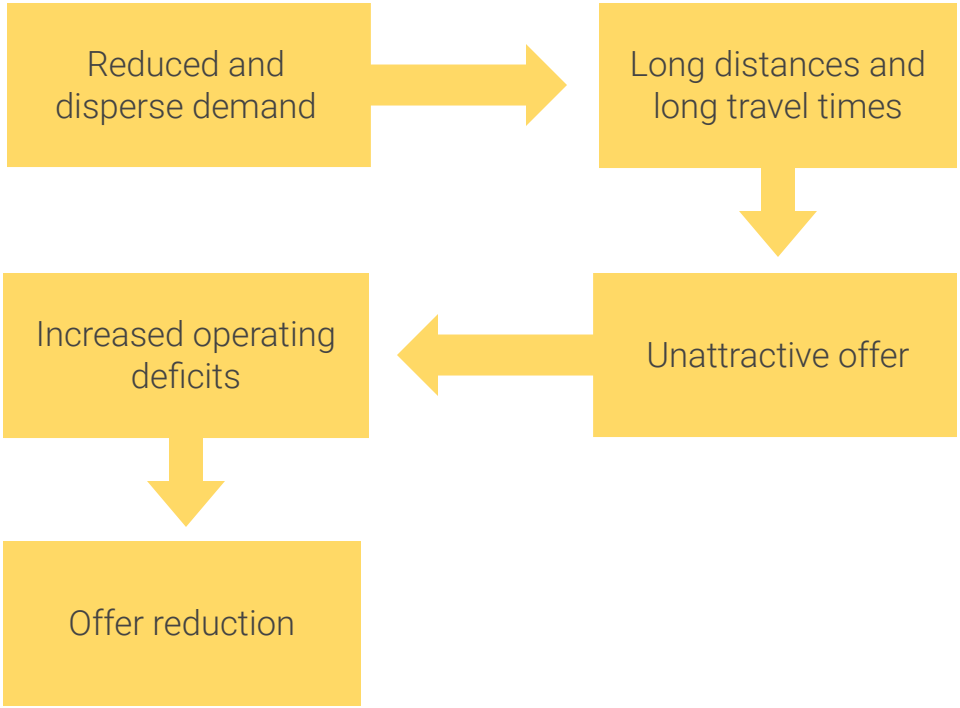
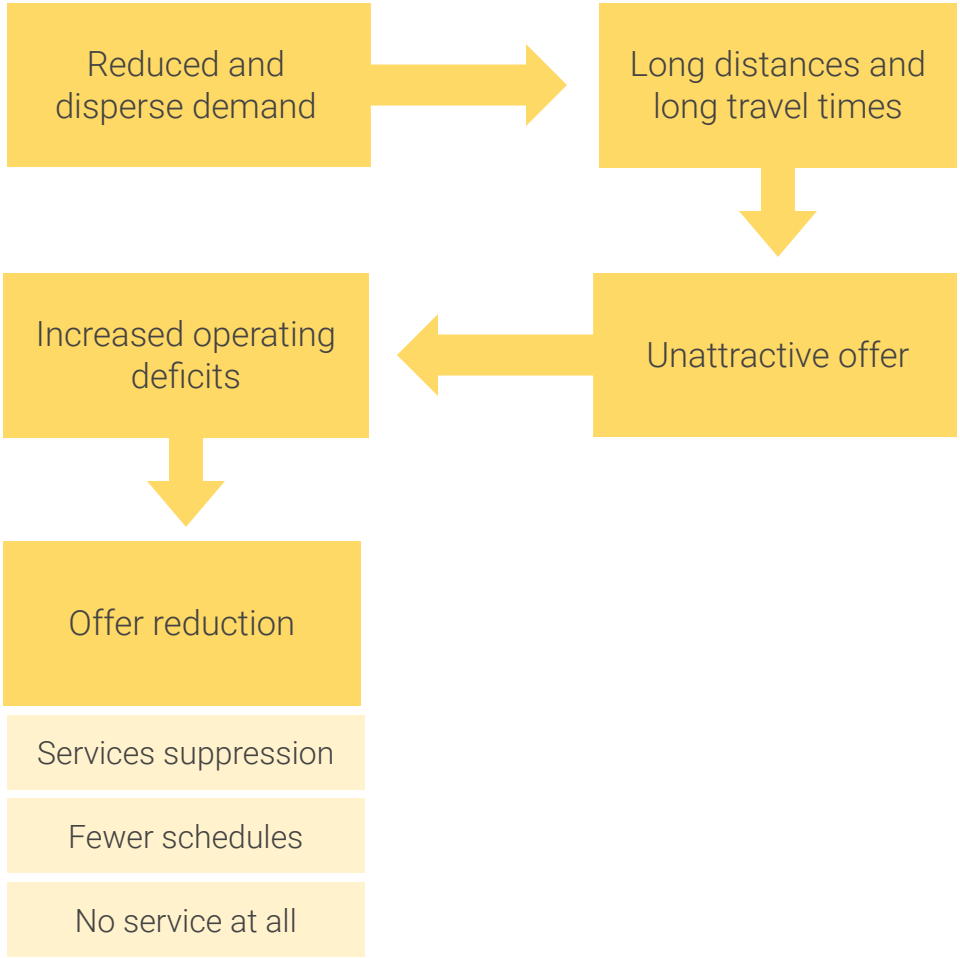


## *In **rural areas**...*



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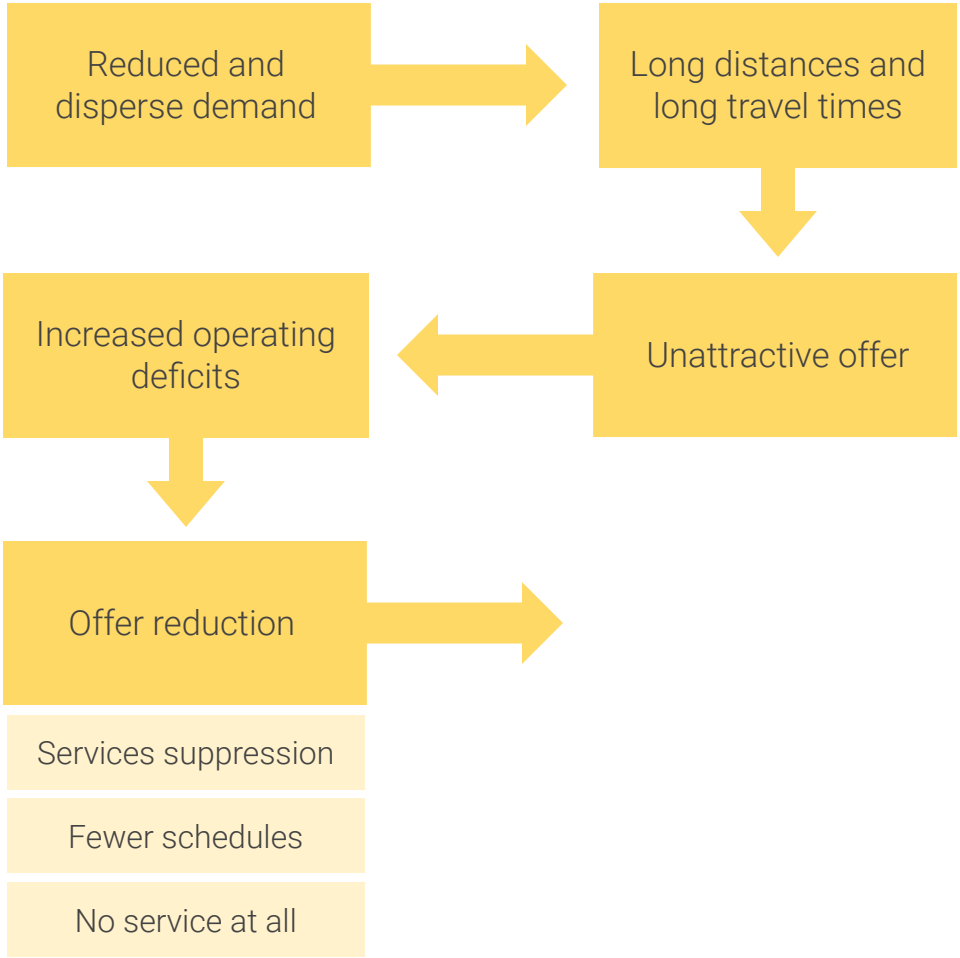

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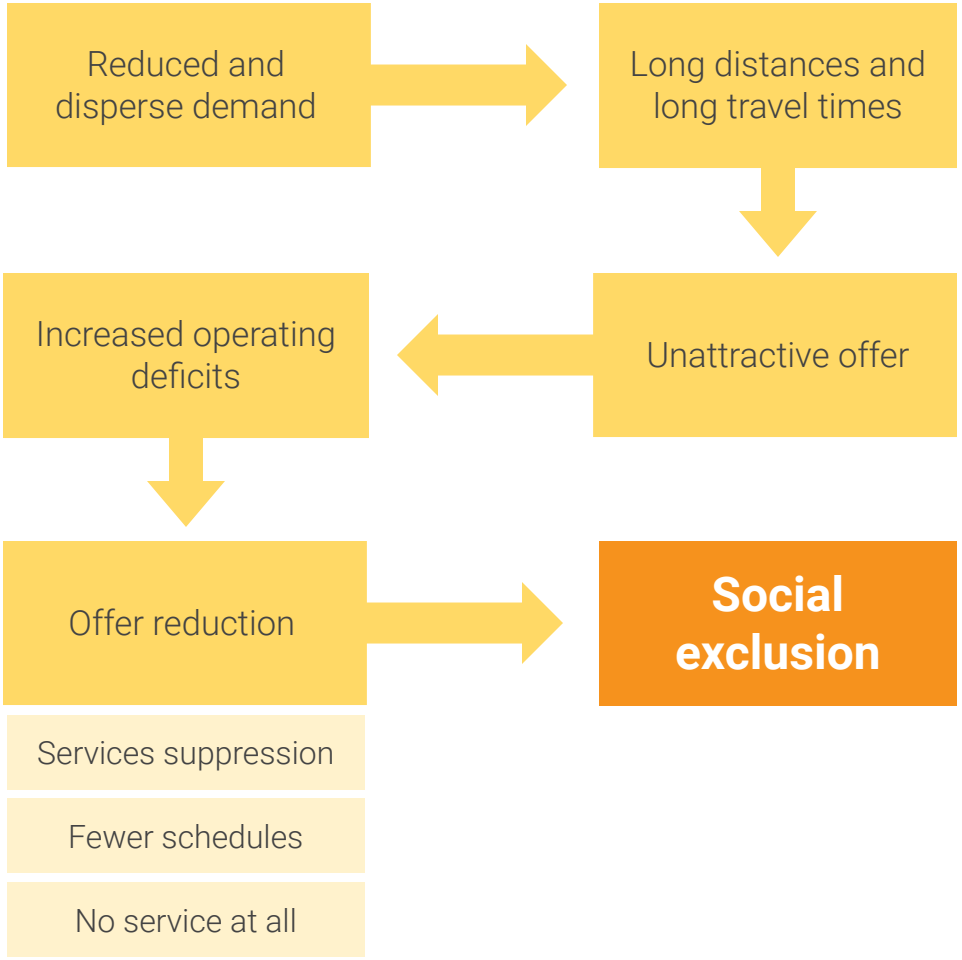

# *How to respond to demand when it is decreasing?*

“

# *How to create sustainable offer in this case?*

“

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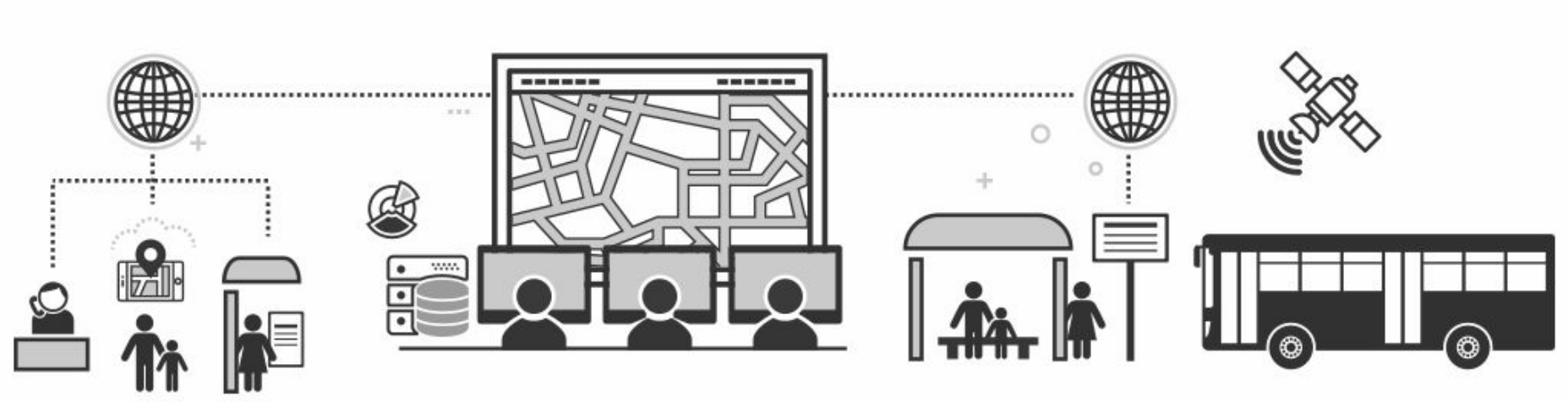
# Demand-responsive transport

An innovative solution

# What is DRT?



# Why DRT in rural areas?

- Adapts to the **existing demand**
- Ensures the **necessary mobility**
- Is **effective**, **efficient** and more **sustainable**
- Contributes to the **economical and social development**

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# Implemented examples

Norway and Portugal

# Flexx @ Norway

## Østfold

284,962 inhabitants

3,887 km$^2$

~68 inhabitants per km$^2$

- Complements the regular bus service;
- No users, no trip;
- Semi-fixed routes *(until 2 km away from the route)*
- Fixed schedule;
- **Minibuses** service the lines;
- Offers **two circular routes**.

**ROUTE [N]** *3 days per week*
*3h-3h*

**ROUTE [N]** *2 days per week*
*3h-3h*

# Flexx @ Norway

## Hedmark

27,397 inhabitants

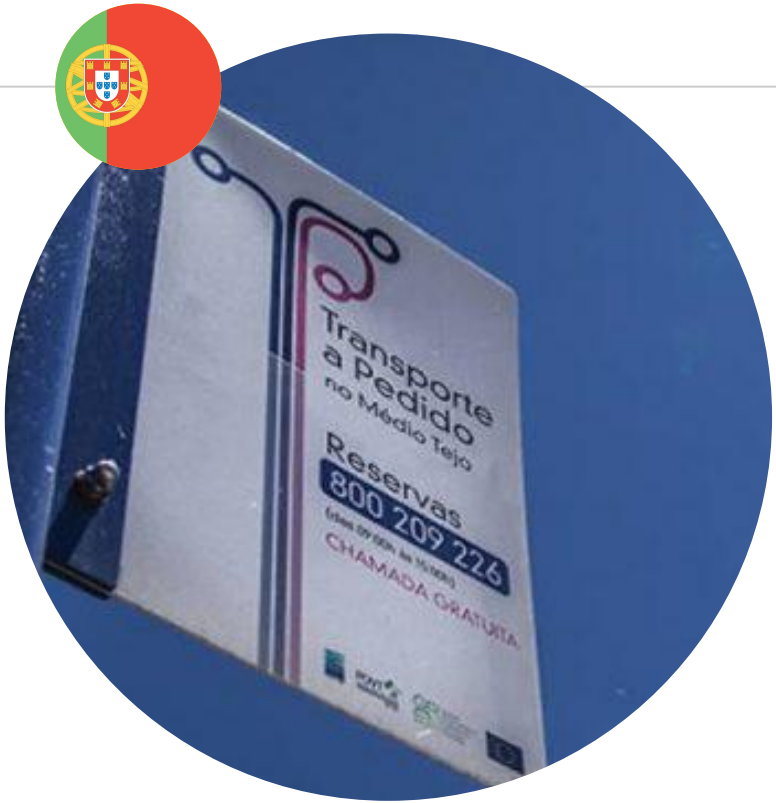
26,084 km$^2$

~8 inhabitants per km$^2$

- Complements the regular bus service;
- No users, no trip;
- Standard fees;
- Fixed stops;
- No fixed route or schedule *(departures every hour)*;
- **Regular cabs** service the lines.

TaP @ Portugal

*Transporte a Pedido*




## Médio Tejo

247,000 inhabitants

3,344 km$^2$

~74 inhabitants per km$^2$

- Complements the existing supply;
- No users, no trip;
- Fixed routes, stops and schedules;
- Guarantees minimum levels of service;
- **Regular cabs** service the lines.
- Ticket price discrimination.

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# Context Analysis

Boticas and Alto Tâmega

# Alto Tâmega Intermunicipal Community

94,143 inhabitants

~3,000 km$^2$

*~33 inhabitants per km$^2$*

*Boticas, Chaves, Montalegre, Ribeira de Pena, Valpaços e Vila Pouca de Aguiar*

# Alto Tâmega Intermunicipal Community

94,143 inhabitants

~3,000 km$^2$

*~33 inhabitants per km$^2$*

*Boticas, Chaves, Montalegre, Ribeira de Pena, Valpaços, Vila Pouca de Aguiar*

# Boticas
# Municipality

5,943 inhabitants

322 km$^2$

*~18 inhabitants per km$^2$*

*Alturas do Barroso e Cerdedo, Ardãos e Bobadela, Beça, Boticas e Granja, Codeçoso, Curros e Fiães do Tâmega, Covas do Barroso, Dornelas, Pinho, Sapiãos, Vilar e Viveiro*

# Boticas Municipality

**28,5%**

> 64 years old

*vs. 20,3% in Portugal*

**501**

vehicles per 1000 inhabitants

*vs. 472 in Portugal*

**18%**

use collective transport

# Taxi

*2.61 vehicles per 1000 inhabitants*

# Bus

*Auto Viação do Tâmega*

## Boticas Municipality

# Bus

*Auto Viação do Tâmega*

# Boticas Municipality

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# Discussion

Critical Analysis and Proposal

# SWOT

## CURRENT SERVICES

| | Helpful | Harmful |
|---|---|---|
| Internal | **S**<br>System covers every settlement<br>Taxis are able to respond to the current weekday demand<br>Public services are organised with the current offer<br>Existence of on-demand services (taxis)<br>Strong flows and high occupancy rates<br>School buses allow other passengers | **W**<br>Less offer on weekends<br>Buses are not prepared for reduced mobility citizens<br>Taxis offer a traditional on-demand system (phone calls)<br>Bus network lack zones and shelters in certain areas<br>Insufficient resources for improving the services<br>Conditions of stakeholders in rural areas are more challenging |
| External | **O**<br>Road network connects every settlement<br>Some services are brought to customers/citizens<br>Legislation enables reorganization of mobility services | **T**<br>Population is decreasing by the day<br>Road network is in poor conditions in certain areas<br>Remarkable rise in costs<br>No competition with other types of services<br>Car-based community |

# SWOT

## CONTEXT

| | Helpful | Harmful |
|---|---|---|
| Internal | **S**<br>Stakeholders are local, and more connected to people<br>Good predictability of demand<br>Solidarity associated with culture<br>Support of authorities | **W**<br>Concentration of business and residents in the seat<br>Mobility based on personal cars<br>Seasonal changes<br>Lack of culture of experimentation |
| External | **O**<br>Road network connects every settlement<br>Bringing services to customers | **T**<br>Population is ageing and descreasing<br>No new market-based services<br>People without a driving license are dependent on the mobility of others |

# SWOT

## MUNICIPALITY

| | Helpful | Harmful |
|---|---|---|
| Internal | **S**<br>Decision-makers are aware of the needs of people | **W**<br>Silo effect between other stakeholders<br>Lack of financial resources |
| External | **O**<br>Intermunicipal collaboration in mobility services<br>Legislation enables reorganization of mobility services | **T**<br>Reducing support and subventions<br>Special groups cannot be taken sufficiently into account<br>Inefficient investment allocation from the state<br>Lack of collaboration and separate visions of stakeholders |

# Main issues

- There is a **lack of offer** during the weekends and summer overall
- Buses and taxis are **not prepared** with the amenities for **reduced mobility users**;
- People without a driving license are **largely dependent** on the mobility of others;
- There is a **lack of culture of experimentation**.

# Example comparison

| | Østfold [NO] | Hedmark [NO] | Médio Tejo [PT] |
|---|---|---|---|
| Fixed routes | Yes | No | Yes |
| Fixed stops | Yes | Yes | Yes |
| Fixed schedule | Yes | No | Yes |
| Vehicles | Minibuses | Taxi | Taxi |

# Example comparison

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# Example comparison

| | Østfold [NO] | Hedmark [NO] | Médio Tejo [PT] | Alto Tâmega [PT] |
|---|---|---|---|---|
| Population | 284,962 | 27,397 | 247,000 | 94,143 |
| Area | 3,887 | 26,084 | 3,344 | 3000 |
| Density | 68 | 8 | 74 | 33 |

# Main features

### Simplicity

It consists in a very basic approach of a DRT system, that would only work on weekends, which have no offer.

### Offer on weekends

It focuses on creating offer on weekends, as there is no bus services and not all the taxis operate.

### Fixed schedules and stops

The target settlements are not big, and their inner streets are not easily accessible by car. Customers would be picked up from the designated bus stops and scheduled time.

### Service vehicles

Depending on the number of passengers and their particular needs the lines would be performed by minibuses or regular taxis.

### Educating the population

Several information sessions would need to be carried out through the municipality, in order to people to take full advantage of this new system.

# Main features

- Prior reservation by phone (24h-1h before the trip);
- Circular routes.

# Some conclusions

**DRT is not holy grail**

DRT by itself will not accomplish all the public transport needs in rural areas.

**We will get there...**

The fact that the new legislation enables the implementation of DRT systems in Portugal is a great advantage for rural municipalities.

# Thanks!

*Any questions?*